PROJECT REPORT

Model Comparison Report

Artificial Intelligence & Machine Learning – Task 2

(Feature Engineering, Model Optimization & Performance Comparison)

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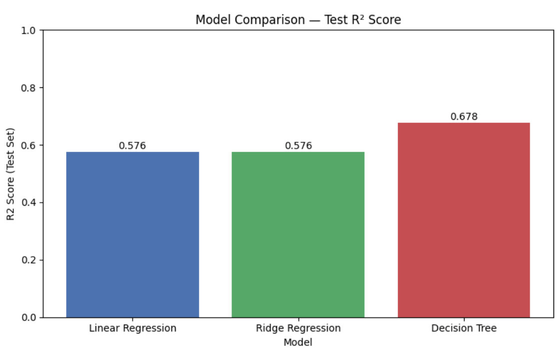
Objective

The objective of this task was to improve the baseline California House Price Prediction model by comparing multiple regression algorithms, evaluating their predictive performance, selecting the most suitable model, and deploying the final solution as an interactive web application.

Methodology

The California Housing Dataset was preprocessed by separating the feature matrix and target variable, followed by an 80:20 train-test split using random_state=42 to ensure reproducibility. Three regression models—Linear Regression, Ridge Regression, and Decision Tree Regressor—were trained and evaluated. Feature scaling was applied only to Linear Regression and Ridge Regression using StandardScaler, while the Decision Tree model was trained on the original feature values. Model performance was assessed using Train RMSE, Test RMSE, Train R^2, Test R^2, and Overfitting Gap (R^2).

Model Performance



Model	Test RMSE	Test R^2
Linear Regression	0.7456	0.5758
Ridge Regression	0.7456	0.5758
Decision Tree Regressor	0.6497	0.6779

The Decision Tree Regressor achieved the highest Test R^2 score (0.6779) and the lowest Test RMSE (0.6497), making it the best-performing model. Consequently, it was selected as the final deployment model.

Model Deployment

The selected Decision Tree Regressor was serialized using Joblib and deployed as a professional Streamlit web application. The application provides three interactive pages: House Price Prediction, where users enter the eight California housing features to obtain real-time predictions; Model Analytics, which compares the performance of Linear Regression, Ridge Regression, and Decision Tree Regressor using interactive Plotly visualizations; and About Developer, which presents the project overview and developer profile.

Conclusion

This project successfully demonstrates an end-to-end machine learning workflow, including data preprocessing, model training, evaluation, comparison, selection, and deployment. By integrating predictive analytics with an interactive Streamlit dashboard, the application provides an efficient and user-friendly solution for California house price prediction while showcasing practical machine learning deployment skills.